

Peripheral Θ -classes and forbidden partial cube-minors of daisy cubes

Guangfu Wang^{a,*}

^aSchool of Mathematics and Information Sciences, Yantai University, Yantai, 264005, Shandong Province, China

Abstract

Daisy cubes are partial cubes whose vertices can be represented by a down-set of a Boolean lattice. This paper gives a label-free characterization: a finite partial cube is a daisy cube if and only if every Djoković–Winkler Θ -class is peripheral. The proof orients each Θ -class toward a peripheral halfspace and shows that the resulting Θ -coordinate labels are closed downward. The characterization turns recognition into a condition on the halfspace structure and gives an exact obstruction formulation: the minimal forbidden pc-minors for daisy cubes are precisely the pc-minor-minimal partial cubes containing a non-peripheral Θ -class. We also give an infinite product family of such obstructions. For all $r \geq 2$ and $s \geq 1$, the graph obtained from $P_3^{\square r} \square Q_s$ by deleting the two opposite corners is a minimal forbidden partial cube-minor for the class of daisy cubes.

Keywords: Daisy cube, Djoković–Winkler relation, peripheral Θ -class, partial cube-minor, forbidden minor

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1. Introduction

All graphs considered here are finite, simple, and connected, unless stated otherwise. For a graph G , let $V(G)$ and $E(G)$ denote its vertex set and edge set. The distance between $u, v \in V(G)$ is denoted by $d_G(u, v)$, or by $d(u, v)$ when the graph is clear. If $S \subseteq V(G)$, then $G[S]$ is the subgraph induced by

*Corresponding author.

Email address: gfwang@ytu.edu.cn (Guangfu Wang)

S . The path on k vertices is denoted by P_k , and K_1 denotes the one-vertex graph.

Partial cubes are graphs whose shortest-path metric is realizable as Hamming distance on binary strings. Equivalently, they are the graphs that occur as isometric subgraphs of hypercubes. Their basic coordinate structure is encoded by the Djoković–Winkler relation: in a partial cube, this relation partitions the edge set into equivalence classes, called Θ -classes. Each Θ -class behaves as one hypercube coordinate and determines two complementary convex halfspaces [5, 8, 15]. This cut structure is one of the main reasons why partial cubes admit a rich theory of convexity, Cartesian products, median-type phenomena, and minor operations adapted to isometric hypercube embeddings.

Daisy cubes were introduced by Klavžar and Mollard [10]. For $X \subseteq \{0, 1\}^n$, the daisy cube generated by X is the subgraph of Q_n induced by the union of all intervals from 0^n to vertices of X ; equivalently, its vertex set is $\downarrow X$, the coordinatewise down-set generated by X . Daisy cubes therefore form a natural down-set subclass of partial cubes. They include hypercubes and several familiar hypercube-like families, such as Fibonacci cubes, Lucas cubes, and bipartite wheels [9, 10, 12].

The theory has developed in several directions. Taranenko [13] characterized daisy cubes by a special expansion procedure and introduced daisy graphs of rooted graphs. Vesel [14] studied proper embeddings and obtained a linear-time proper-embedding algorithm for graphs known to be daisy cubes. Dravec and Taranenko [6] extended the theme to daisy Hamming graphs. Polynomial characterizations and identities for daisy cubes have also been investigated, most recently by Zheng, Xie, and Xu [17]. In chemical graph theory, daisy cubes occur as resonance graphs of suitable plane bipartite graphs [1], with subsequent work on decomposition and τ -graph structure [2, 3].

The down-set definition is useful, but it is not an intrinsic graph-theoretic description. To use it on an abstract graph, one must find an isometric embedding into a hypercube, choose a vertex to serve as 0^n , and orient all coordinates so that the resulting labels form a down-set. A structural criterion that avoids these choices is therefore desirable. Xie and Xu [16] recently proved, within the class of finite median graphs, that being a daisy cube is equivalent to every Θ -class being peripheral, and they also emphasized the forbidden pc-minor problem for daisy cubes. The first aim of the present paper is to remove the median assumption from this peripheral characteriza-

tion.

We call a Θ -class peripheral when one of its two halfspaces consists entirely of vertices incident with an edge from that class. Our first main result proves that a finite partial cube is a daisy cube exactly when all its Θ -classes are peripheral. Directly choose, for each class, a peripheral halfspace as the 1-side of the coordinate. Then every vertex with coordinate value 1 has a neighbor obtained by changing only that coordinate to 0. Repeating this operation shows that the set of Θ -coordinate labels is closed downward.

This characterization also clarifies the obstruction problem. Since daisy cubes are closed under restrictions to halfspaces and contractions of Θ -classes, one may ask for the minimal forbidden partial cube-minors. The theorem shows that the obstruction set is exactly the set of pc-minor-minimal partial cubes that possess a non-peripheral Θ -class. This statement is not a complete enumeration, but it reduces the classification target to a concrete halfspace condition.

The Cartesian product of graphs G and H is denoted by $G \square H$ [7, 8]. Its vertex set is $V(G) \times V(H)$, and two vertices (g, h) and (g', h') are adjacent precisely when either $g = g'$ and $hh' \in E(H)$, or $h = h'$ and $gg' \in E(G)$. We write $G^{\square r}$ for the r -fold Cartesian product of G and use the convention $G^{\square 0} = K_1$.

Our second main result supplies a new infinite family of minimal obstructions. Let

$$B_{r,s} = P_3^{\square r} \square Q_s,$$

with vertex set $\{0, 1, 2\}^r \times \{0, 1\}^s$. For $r \geq 2$ and $s \geq 1$, let $M_{r,s}$ be obtained from $B_{r,s}$ by deleting the two opposite corners

$$\alpha = (0, \dots, 0; 0, \dots, 0), \quad \beta = (2, \dots, 2; 1, \dots, 1).$$

We prove that every $M_{r,s}$ is a minimal forbidden pc-minor for daisy cubes. This extends the previously recorded two-corner deletion examples from the cases with no P_3 factor or one P_3 factor to higher products of P_3 factors.

2. Notation and preliminaries

For $n \geq 0$, the hypercube Q_n has vertex set $\{0, 1\}^n$; two binary strings are adjacent if they differ in exactly one coordinate. We set $Q_0 = K_1$. A subgraph H of a graph G is *isometric* if $d_H(x, y) = d_G(x, y)$ for all $x, y \in$

$V(H)$. A *partial cube* is a graph isomorphic to an isometric subgraph of a hypercube.

For edges $ab, xy \in E(G)$ of a connected graph G , the Djoković–Winkler relation is

$$ab \Theta_G xy \iff d(a, x) + d(b, y) \neq d(a, y) + d(b, x).$$

When no confusion is possible, we write Θ instead of Θ_G . A connected bipartite graph is a partial cube if and only if Θ is an equivalence relation on its edge set [5, 15]. The equivalence classes are called Θ -classes. In a partial cube, the number of Θ -classes is the isometric dimension, that is, the dimension of the standard hypercube embedding.

Let G be a partial cube and let $ab \in E(G)$. Define

$$\begin{aligned} W_{ab} &= \{x \in V(G) : d(a, x) < d(b, x)\}, \\ W_{ba} &= \{x \in V(G) : d(b, x) < d(a, x)\}. \end{aligned}$$

These sets are complementary convex halfspaces of G . The Θ -class containing ab is

$$F_{ab} = \{xy \in E(G) : x \in W_{ab} \text{ and } y \in W_{ba}\}.$$

For the same oriented edge ab , put

$$U_{ab} = \{x \in W_{ab} : x \text{ is incident with an edge of } F_{ab}\},$$

and define U_{ba} analogously. Equivalently, if θ is a Θ -class and W is one of its two halfspaces, then

$$U(\theta, W) = \{x \in W : x \text{ is incident with an edge of } \theta\}.$$

Thus $U_{ab} = U(F_{ab}, W_{ab})$.

Definition 2.1. A Θ -class θ of a partial cube is *peripheral* if $U(\theta, W) = W$ for at least one of the two halfspaces W determined by θ .

This is the same terminology as in [16]; here no median-graph assumption is imposed.

For a partial cube G with m Θ -classes, choose for the i th class one halfspace W_i^0 and denote the other one by W_i^1 . The standard Θ -coordinate label of a vertex v is the binary string $\lambda(v) \in \{0, 1\}^m$ given by

$$\lambda(v)_i = \varepsilon \iff v \in W_i^\varepsilon, \quad \varepsilon \in \{0, 1\}.$$

The map λ is an isometric embedding into Q_m . Moreover, an edge of G belongs to the i th Θ -class exactly when the labels of its endpoints differ only in coordinate i .

The coordinatewise order on $\{0, 1\}^n$ is denoted by $\leq$. Thus, for $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$, we have $y \leq x$ if $y_i \leq x_i$ for all i . For $X \subseteq \{0, 1\}^n$, write

$$\downarrow X = \{y \in \{0, 1\}^n : y \leq x \text{ for some } x \in X\}.$$

A nonempty set $D \subseteq \{0, 1\}^n$ is a *down-set* if $D = \downarrow D$.

Definition 2.2. A graph is a *daisy cube* if it is isomorphic to $Q_n[D]$ for some $n \geq 0$ and some nonempty down-set $D \subseteq \{0, 1\}^n$. Equivalently, it is isomorphic to $Q_n[\downarrow X]$ for some $X \subseteq \{0, 1\}^n$.

A *restriction* of a partial cube G with respect to a Θ -class is the subgraph induced by one of the two halfspaces of that class. A *contraction* of G with respect to a Θ -class is obtained by contracting all edges of that class and then suppressing loops and parallel edges. A *partial cube-minor*, abbreviated *pc-minor*, is any graph obtained by a finite sequence of restrictions and contractions. Such a pc-minor is *proper* if it is not isomorphic to the original graph. Restrictions and contractions preserve partial cubes [4, 11], and daisy cubes are pc-minor-closed [10, 13, 16].

Lemma 2.3. *Let H be an isometric subgraph of a partial cube G . If $e, f \in E(H)$, then*

$$e \Theta_H f \iff e \Theta_G f.$$

Consequently, each Θ -class of H is the nonempty intersection of $E(H)$ with a Θ -class of G , and the halfspaces of H are obtained by intersecting the corresponding halfspaces of G with $V(H)$.

Proof. The relation Θ is defined only in terms of the four distances among the endpoints of the two edges. These distances are the same in H and in G because H is isometric in G . The statements about Θ -classes and halfspaces follow immediately from the definitions. $\square$

3. A peripheral characterization of daisy cubes

Theorem 3.1. *A finite partial cube is a daisy cube if and only if every Θ -class is peripheral.*

Proof. Suppose first that G is a daisy cube. Identify G with $Q_n[D]$, where D is a nonempty down-set in $\{0, 1\}^n$. Consider a nonempty coordinate class, say the class corresponding to coordinate i . If $x \in D$ and $x_i = 1$, let x' be obtained from x by changing the i th coordinate from 1 to 0 and leaving all other coordinates unchanged. Then $x' \leq x$, so $x' \in D$. The vertices x and x' are adjacent in G , and their edge lies in the coordinate- i class. Hence every vertex in the halfspace $\{x \in D : x_i = 1\}$ is incident with an edge of this class. This halfspace is equal to the corresponding U -set, and the class is peripheral.

Conversely, assume that every Θ -class of the partial cube G is peripheral. Let $\theta_1, \dots, \theta_m$ be the Θ -classes. For each i , choose a peripheral halfspace of θ_i and denote it by W_i^1 ; denote the opposite halfspace by W_i^0 . If both halfspaces are peripheral, choose either one as W_i^1 . Define the standard Θ -coordinate embedding

$$\lambda : V(G) \longrightarrow \{0, 1\}^m, \quad \lambda(v)_i = \varepsilon \iff v \in W_i^\varepsilon.$$

For $\varepsilon \in \{0, 1\}$, write $U_i^\varepsilon = U(\theta_i, W_i^\varepsilon)$. By the choice of the orientation, $U_i^1 = W_i^1$ for all i .

We prove that $\lambda(V(G))$ is a down-set. Let $v \in V(G)$ and suppose that $\lambda(v)_i = 1$. Then $v \in W_i^1 = U_i^1$, so v is incident with an edge vv' of the class θ_i whose other endpoint lies in W_i^0 . In the standard Θ -coordinate embedding, the labels of the endpoints of an edge in θ_i differ exactly in coordinate i . Therefore $\lambda(v')$ is obtained from $\lambda(v)$ by changing the i th coordinate from 1 to 0 and changing no other coordinate.

Now take any $y \leq \lambda(v)$. Starting at v , apply the preceding coordinate-lowering step successively to each coordinate in which $\lambda(v)$ has value 1 and y has value 0. At each intermediate vertex, every coordinate that still has to be lowered is still equal to 1, so the same argument applies again. The final vertex has label y . Thus every binary string below a label in $\lambda(V(G))$ again belongs to $\lambda(V(G))$, and hence $\lambda(V(G))$ is a nonempty down-set of $\{0, 1\}^m$.

Since λ is an isometric embedding, two vertices of G are adjacent exactly when their labels differ in one coordinate. Consequently, G is isomorphic to the subgraph of Q_m induced by the down-set $\lambda(V(G))$. Hence G is a daisy cube. $\square$

Let $\mathcal{O}_D$ be the set of all partial cubes H such that H is not a daisy cube and every proper pc-minor of H is a daisy cube. These graphs are the minimal forbidden pc-minors for the class of daisy cubes.

Corollary 3.2. *For a finite partial cube G ,*

$$G \text{ is a daisy cube} \iff G \text{ has no pc-minor isomorphic to a member of } \mathcal{O}_D.$$

Moreover, $\mathcal{O}_D$ is exactly the set of pc-minor-minimal partial cubes that contain a non-peripheral Θ -class.

Proof. Because daisy cubes are pc-minor-closed, a daisy cube cannot have a pc-minor in $\mathcal{O}_D$. Conversely, if G is not a daisy cube, then among all non-daisy pc-minors of G choose one with minimum $|V| + |E|$. This chosen pc-minor belongs to $\mathcal{O}_D$. It follows directly from Theorem 3.1 that, within finite partial cubes, being non-daisy is equivalent to having at least one non-peripheral Θ -class. $\square$

Remark 3.3. Corollary 3.2 is an exact obstruction-set formulation, not a classification of all members of $\mathcal{O}_D$. A full forbidden pc-minor theorem for daisy cubes is therefore equivalent to enumerating the pc-minor-minimal partial cubes with a non-peripheral Θ -class.

4. A new infinite family of minimal obstructions

For integers $r, s \geq 0$, set

$$B_{r,s} = P_3^{\square r} \square Q_s.$$

We use the coordinate model

$$V(B_{r,s}) = \{0, 1, 2\}^r \times \{0, 1\}^s,$$

in which two vertices are adjacent if exactly one coordinate changes by 1 and all other coordinates remain fixed. If $r = 0$ or $s = 0$, the corresponding block of coordinates is omitted. For $r \geq 2$ and $s \geq 1$, define

$$\alpha = (0, \dots, 0; 0, \dots, 0), \quad \beta = (2, \dots, 2; 1, \dots, 1),$$

and let

$$M_{r,s} = B_{r,s} - \{\alpha, \beta\}.$$

Here $G - S$ denotes the induced subgraph $G[V(G) \setminus S]$.

We first record two elementary facts about products of paths. For

$$B = P_{\ell_1+1} \square \cdots \square P_{\ell_t+1},$$

with vertex set $\prod_{i=1}^t \{0, 1, \dots, \ell_i\}$, a *corner* is a vertex whose i th coordinate is either 0 or ℓ_i for every i . The corners $(0, \dots, 0)$ and $(\ell_1, \dots, \ell_t)$ are called opposite.

Lemma 4.1. *Let $B = P_{\ell_1+1} \square \cdots \square P_{\ell_t+1}$, where each $\ell_i \geq 1$. Put*

$$a = (0, \dots, 0), \quad b = (\ell_1, \dots, \ell_t), \quad L = \ell_1 + \cdots + \ell_t.$$

If $L \geq 3$, then $B - \{a, b\}$ is an isometric subgraph of B .

Proof. For $z = (z_1, \dots, z_t)$, set $h(z) = z_1 + \cdots + z_t$. Let $x, y \in V(B) \setminus \{a, b\}$. We construct a shortest x, y -path in B that avoids a and b . In a product of paths, every shortest path is obtained by changing coordinates monotonically from their values in x to their values in y .

Assume that the current vertex is $z \neq y$. If $h(z) = 1$, choose a coordinate in which the next monotone move toward y increases the coordinate. Such a coordinate must exist; otherwise every remaining move toward y would be decreasing, and since z has exactly one unit of total coordinate sum, this would force $y = a$, contrary to $y \neq a$. Similarly, if $h(z) = L - 1$, choose a decreasing move toward y ; such a move exists unless $y = b$. If $1 < h(z) < L - 1$, choose any monotone move toward y .

Each chosen move decreases the distance to y by one. The process therefore stops after exactly $\sum_i |x_i - y_i| = d_B(x, y)$ steps. The construction keeps every intermediate vertex at a value of h between 1 and $L - 1$, so neither a nor b is used. Hence $d_{B-\{a,b\}}(x, y) = d_B(x, y)$ for all remaining vertices x and y . $\square$

Lemma 4.2. *For all $p, q \geq 0$, the product $P_3^{\square p} \square Q_q$ is a daisy cube. If one corner is deleted and the remaining graph is nonempty, then the remaining graph is also a daisy cube.*

Proof. The path P_3 is isomorphic to the subgraph of Q_2 induced by the down-set $\{00, 10, 01\}$. The hypercube Q_q is induced by the down-set $\{0, 1\}^q$. The Cartesian product of two daisy cubes is a daisy cube: if the factors are induced by down-sets $D_1 \subseteq \{0, 1\}^m$ and $D_2 \subseteq \{0, 1\}^n$, then their product is induced by the down-set $D_1 \times D_2 \subseteq \{0, 1\}^{m+n}$. Hence $P_3^{\square p} \square Q_q$ is a daisy cube.

Let c be a corner of $P_3^{\square p} \square Q_q$. In each P_3 factor, use the embedding $P_3 \cong Q_2[\{00, 10, 01\}]$ so that the endpoint appearing in c receives one of the two maximal labels, 10 or 01. In each hypercube coordinate, orient the coordinate so that the endpoint appearing in c receives label 1. In the resulting product embedding, c is a maximal element of the down-set. Deleting a maximal element from a finite down-set leaves a down-set. Therefore the one-corner deletion is again a daisy cube whenever it is nonempty. $\square$

Theorem 4.3. *For every $r \geq 2$ and $s \geq 1$, the graph $M_{r,s}$ is a minimal forbidden pc-minor for the class of daisy cubes.*

Proof. The graph $B_{r,s}$ is a Cartesian product of paths, hence a partial cube. The distance between α and β in $B_{r,s}$ is $2r + s \geq 5$. By Lemma 4.1, $M_{r,s}$ is an isometric subgraph of $B_{r,s}$, and so $M_{r,s}$ is a partial cube.

We show next that $M_{r,s}$ is not a daisy cube. Fix one coordinate j from the $\{0, 1\}^s$ block, and let θ be the corresponding Θ -class of $M_{r,s}$. Let α_j^+ be the vertex obtained from α by changing this coordinate from 0 to 1, and let β_j^- be the vertex obtained from β by changing this coordinate from 1 to 0. Both vertices belong to $M_{r,s}$.

In $B_{r,s}$, the only edge of the j th cube-coordinate class incident with α_j^+ joins α_j^+ to α , and the only such edge incident with β_j^- joins β_j^- to β . These two neighbors have been deleted. Since $M_{r,s}$ is isometric in $B_{r,s}$, Lemma 2.3 shows that the Θ -classes and halfspaces of $M_{r,s}$ are inherited from $B_{r,s}$. Thus one halfspace of θ contains the vertex α_j^+ , which is not incident with any edge of θ in $M_{r,s}$, and the other halfspace contains the vertex β_j^- , which is also not incident with any edge of θ in $M_{r,s}$. Hence neither halfspace is equal to its U -set. The class θ is non-peripheral, so $M_{r,s}$ is not a daisy cube by Theorem 3.1.

It remains to prove minimality. It suffices to verify that every elementary restriction and every elementary contraction of $M_{r,s}$ is a daisy cube. Then any further pc-minor is a daisy cube because the class of daisy cubes is pc-minor-closed.

The Θ -classes of $M_{r,s}$ are inherited from the coordinate classes of $B_{r,s}$. There are three types to consider.

First, let θ be a class belonging to one of the s hypercube coordinates. Restricting to either halfspace fixes that coordinate. The deleted vertices α and β lie in opposite halfspaces, so each restriction is isomorphic to $B_{r,s-1}$ with one corner deleted. Contracting this class collapses the chosen cube coordinate and gives the full product $B_{r,s-1}$: the contraction class containing

α has the surviving representative α_j^+ , and the contraction class containing β has the surviving representative β_j^- . By Lemma 4.2, all these graphs are daisy cubes.

Second, consider a lower class in a P_3 coordinate, namely the class consisting of edges between levels 0 and 1 in that coordinate. One restriction fixes the coordinate at level 0; it is isomorphic to $B_{r-1,s}$ with the corner corresponding to α deleted. The other restriction keeps levels 1 and 2; since this two-level factor is a $P_2 \cong Q_1$, the restriction is isomorphic to $B_{r-1,s+1}$ with the corner corresponding to β deleted. Contracting the lower class turns the chosen P_3 factor into a P_2 factor. The image of α is present because its contraction class contains the surviving vertex obtained from α by changing the chosen coordinate from 0 to 1, whereas the image of β remains deleted. Hence the contraction is isomorphic to $B_{r-1,s+1}$ with one corner deleted. Lemma 4.2 again shows that all three elementary pc-minors in this case are daisy cubes.

Third, consider an upper class in a P_3 coordinate, consisting of the edges between levels 1 and 2. This is symmetric to the lower-class case. The two restrictions are isomorphic to $B_{r-1,s+1}$ with one corner deleted and to $B_{r-1,s}$ with one corner deleted. The contraction is isomorphic to $B_{r-1,s+1}$ with one corner deleted; now the contraction class containing β has a surviving representative, while the image of α is the deleted corner. These graphs are daisy cubes by Lemma 4.2.

Every elementary pc-minor of $M_{r,s}$ is therefore a daisy cube. Since $M_{r,s}$ itself is not a daisy cube, it is a minimal forbidden pc-minor for the class of daisy cubes. $\square$

Adopt the notation from [17], let Q_s^{--} denote the graph obtained from Q_s by deleting two opposite vertices; that is, $Q_s^{--} = Q_s - \{0^s, 1^s\}$.

Remark 4.4. When the same two-corner deletion construction is applied to Q_s , one obtains Q_s^{--} , the hypercube with two opposite vertices deleted. Similarly, applying the construction to $P_3 \square Q_s$ gives the previously recorded obstruction pattern H_{s+2}^{--} , up to notation. Theorem 4.3 treats the higher-product case $P_3^{\square r} \square Q_s$ with $r \geq 2$.

5. Concluding remarks

Theorem 3.1 replaces the labelled Boolean-down-set definition of a daisy cube by an intrinsic condition on Θ -classes. Equivalently, a finite partial cube is non-daisy precisely when some Θ -class has, on each of its two sides,

a vertex not incident with that class. This criterion answers the need for a label-free test and gives a precise target for the forbidden pc-minor problem: classify the pc-minor-minimal partial cubes with a non-peripheral Θ -class.

The graphs $M_{r,s}$ show that the obstruction set is infinite and includes higher-dimensional product configurations beyond the initial examples Q_s^{--} and H_n^{--} . A complete enumeration of $\mathcal{O}_D$ remains open.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used OpenAI's ChatGPT to assist with language editing, and structural organization. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data availability

No data were used for the research described in the article.

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